好的，我仔细看了你新上传的 **2025 JAN** 考卷，这份考卷的题型和 Tips 非常吻合。

我根据 **Tips** + **2025 JAN** + **之前的 Paper**，重新帮你生成一份 **预测考卷 Format**，题型、配分、考点都围绕 Tips，但**不会跟任何一份考卷一模一样**，避免你只背答案。

AMCS1113 COMPUTER ARCHITECTURE
PREDICTED FINAL EXAMINATION PAPER (Based on Tips + 2025 JAN + Past Papers)

Time: 2 Hours
Answer ALL questions. All questions carry equal marks.
Total: 100 marks

Question 1 (25 marks)

a) Perform the following conversions. Show all working steps clearly. If the operation(s) is invalid, explain the reason. (5 marks)
(i) Convert `$247_8$` to base-10 number. (2 marks)
(ii) Convert `$1011010.101_2$` to base-16 number. (3 marks)

b) Perform the following arithmetic operations. Show all working steps clearly. (6 marks)
(i) `$11010110_2 + 101101_2$` (3 marks)
(ii) `$2A7_{16} \times 3F_{16}$` (3 marks)

c) IEEE 754 Single Precision Format (7 marks)
Given a decimal number `-45.5$_{10}$`, show how this number can be represented in the **IEEE 754 single precision notation** using **Excess-127** notation. You are required to show your working steps.

d) Two's Complement (7 marks)
Assume an 8-bit system is applied. Solve the following operation using the **Two's Complement** method.

`-10 + (-78)`

(i) Provide the result in signed decimal form. (4 marks)
(ii) Identify whether a carry occurs. (1 mark)
(iii) Identify whether an overflow occurs. (1 mark)
(iv) Explain why the carry is discarded in 2's complement addition. (1 mark)

(温馨提示, 这个我给我自己看的罢了)
Fight: 9+6+5+5 = 25/25

Question 2 (25 marks)

a) RGB Pixel Representation (5 marks)
A 1-pixel image is stored in memory starting at address `0x6009`. The RGB values for the pixel are as follows:

Pixel 1: Mixed colour (153, 15, 75)

Using hexadecimal format, show the value stored at each memory address for this pixel in a table:

Memory Address	Stored Value

b) Little Man Computer (LMC) Operation (10 marks)
Using the Little Man Computer (LMC) model, consider the following instructions stored at the specified memory locations:

- Program counter: 30
- Value in memory location 30: 580 (LOAD)
- Value in memory location 31: 281 (SUBTRACT)
- Value in memory location 32: 380 (STORE)
- Value in memory location 80: 055
- Value in memory location 81: 010

Trace and show the changes in the contents of the **Instruction Register (IR)**, **Program Counter (PC)**, **Memory Address Register (MAR)**, **Memory Data Register (MDR)**, and

Accumulator (A) during the execution of instructions at locations **31 and 32**. (10 marks)

c) CPU Components (6 marks)
Describe the role of the following CPU components:
(i) Control Unit (CU) (2 marks)
(ii) Register (2 marks)
(iii) Arithmetic Logic Unit (ALU) (2 marks)

d) Bus Types (4 marks)
Differentiate between **Point-to-Point Bus** and **Multipoint Bus** in a table format. Provide ONE example for each.

(温馨提示, 这个我给我自己看的罢了)
Fight: 3+10+4+4 = 21/25

Question 3 (25 marks)

a) IPO(S) Framework (9 marks)
In the context of the computer Input-Process-Output (IPO) framework, provide ONE (1) real-world example to illustrate the four main operations of the IPO framework.

b) Cache Memory (8 marks)
(i) Explain the term **cache hit** with the aid of a diagram. (4 marks)
(ii) Describe what will happen when a cache memory is full. (4 marks)

c) Virtual Memory (8 marks)
A computer has limited physical memory (RAM) but needs to run a large program. Explain how **virtual memory** allows the program to run.

(温馨提示, 这个我给我自己看的罢了)
Fight: 9+4+5 = 18/25

Question 4 (25 marks)

a) EAT Calculation (5 marks)
Given that:
- TLB Access Time: 10 ns
- Memory Access Time: 80 ns
- TLB Hit Rate: 90%

Calculate the **EAT** and present the final result in nanoseconds (ns).

b) Page Number Calculation (8 marks)
A paging system uses:
- Logical address size = 20 bits
- Page size = 4 KB
- Logical address = `95A52H`
- Page table entry:

Page No	Frame No
95H	3A7FH

(i) Determine the number of bits used for page number and offset. (2 marks)
(ii) Determine the page number and offset. (2 marks)
(iii) Determine the frame number. (1 mark)
(iv) Construct the physical address. (2 marks)
(v) Explain why the offset remains unchanged during address translation. (1 mark)

c) Volatile vs Non-Volatile Memory (6 marks)
A computer suddenly loses power. Explain how **volatile memory** and **non-volatile memory** affect:

(i) Data retention (3 marks)
(ii) Data loss due to some error (3 marks)

d) TLB Advantage and Disadvantage (6 marks)
(i) State TWO (2) advantages of TLB. (3 marks)
(ii) State TWO (2) disadvantages/limitations of TLB. (3 marks)

(温馨提示, 这个我给我自己看的罢了)
Fight: 6+4+6 = 18/25

📊 配分总结

Question	Topic	Marks
Q1	Number Systems + IEEE 754 + Two's Complement	25
Q2	RGB + LMC + CPU Components + Bus Types	25
Q3	IPO + Cache + Virtual Memory	25
Q4	EAT + Page Number + Volatile/Non-Volatile + TLB	25
Total		**100**

✅ 对应 Tips 检查

Tips	出现在	题型
Chapter 1: Subtraction & Multiplication	Q1b	Calculation
Chapter 1: Arithmetic Operations	Q1b	Calculation
Chapter 2: 2's Complement	Q1d	Calculation
Chapter 2: Carry/Overflow	Q1d	Calculation

Chapter 2: IEEE 754	Q1c	Calculation
Chapter 2: RGB Hex	Q2a	Calculation
Chapter 3: LMC Operation (10 marks)	Q2b	Calculation/Tracing
Chapter 3: CPU Components (CU, Register, ALU)	Q2c	Theory
Chapter 3: Point-to-Point vs Multipoint Bus	Q2d	Theory
Chapter 3: IPO + Real Example	Q3a	Theory
Chapter 4: EAT Calculation	Q4a	Calculation
Chapter 4: Page Number	Q4b	Calculation
Chapter 4: Volatile vs Non-Volatile	Q4c	Theory (Scenario)
Chapter 4: TLB Advantage/Disadvantage	Q4d	Theory
Chapter 4: Cache Hit/Miss	Q3b	Theory
Chapter 4: Virtual Memory (Scenario)	Q3c	Theory

💡 这份预测卷跟 2025 JAN 的对比

项目	2025 JAN	预测卷
Q1a	Base-5, Base-8, Base-16 转换	Base-8, Base-16 转换
Q1b	Binary addition, Hex multiplication	Binary addition, Hex multiplication
Q1c	SEEMMMMM	**IEEE 754**（Tips 说 SEEMMMMM 不出）
Q1d	IEEE 754	**Two's Complement**（Tips 说 2's Complement 出）
Q2a	Point-to-Point vs Multipoint	Point-to-Point vs Multipoint
Q2b	RGB（给 memory table 反推 RGB）	RGB（给 RGB 正推 memory table）
Q2c	LMC（10 marks）	LMC（10 marks）
Q2d	Data/Address/Control Bus	CPU Components + Bus Types
Q3a	Volatile vs Non-Volatile	IPO + Real Example

Q3b	Cache Miss	Cache Hit/Miss
Q3c	Memory Interleaving	Virtual Memory
Q3d	Internal vs External Fragmentation	—
Q3e	EAT Calculation	—
Q4a	Virtual Memory	—
Q4b	Page Table	—
Q4c	TLB Advantages	TLB Advantages
Q4d	TLB Miss vs Page Fault	TLB Disadvantage

💡 这份预测卷的特点

1. **完全根据 Tips 出题**，没有超出范围。
2. **SEEMMMMM 不出**，改成 **IEEE 754**。
3. **RISC/CISC 不出**。
4. **Bus 改成 Point-to-Point vs Multipoint**，不是 Internal/External。
5. **LMC 是 10 marks**，跟 Tips 说的一样。
6. **IPO 跟 2025 OCT Q3b 一样**。
7. **EAT + Page Number** 都在 Q4。
8. **Volatile/Non-Volatile** 是情景题。
9. **TLB 优缺点** 都在 Q4。
10. **Cache Hit/Miss** 和 **Virtual Memory** 都在 Q3。

这份预测卷 **刚好 100 marks**，你可以拿来做模拟练习。